

$$h = 0.50 \text{ m}$$

11 A

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DUNMAN HIGH SCHOOL
2023 H2 PHYSICS (YEAR 6)

period of rotation = 365 days

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{(365 \times 24 \times 60 \times 60)} = 1.99 \times 10^{-7} \text{ rad s}^{-1}$$

$$v = r\omega = (1.50 \times 10^{11})(1.99 \times 10^{-7}) = 2.99 \times 10^4 \text{ m s}^{-1}$$